Enhancing Multimodal Understanding through Integrated Image Embeddings and Large Language Models using Reinforcement Learning

Abstract

This research proposal outlines a method to improve multimodal understanding by combining image embeddings with large language models (LLMs) like Llama-3[11] or GPT-3[12]. We use a mapping network to project CLIP[1] image embeddings into the LLM space, enabling the LLM to generate responses based on both text and images. We apply reinforcement learning techniques, specifically Self-Critical Sequence Training (SCST)[2], for sequence-level optimization. The expected outcomes include advancements in multimodal comprehension, generalization across diverse image-question tasks, and performance comparable to GPT-4's image-question answering capabilities. The method suggests a streamlined architecture with few trainable parameters.

1. Introduction

Multimodal understanding, the ability to comprehend and process information from multiple modalities such as text and images, is a fundamental challenge in artificial intelligence. Recent advancements in CLIP and large language models (LLMs) like GPT-3 have significantly advanced the field by offering powerful tools for processing and generating textual and visual data. However, bridging the semantic gap between these modalities remains a key challenge. In this research proposal, we introduce an architecture to enhance multimodal understanding by integrating image embeddings with LLMs. Our method combines CLIP, a model specialized in image understanding, with pre-trained LLMs such as GPT-3 or Llama-3. By concatenating CLIP embeddings with prompt embeddings from the LLM encoder, we enable the model to generate responses based on a combination of textual prompts and image inputs. A simple mapping network is employed to project CLIP embeddings into the LLM space, facilitating the model's understanding of visual data. Fine-tuning the LLM on image-question answering tasks further enhances its ability to generate meaningful responses. We anticipate that our approach will effectively leverage the semantic features of CLIP and the knowledge encoded within pre-trained LLMs, potentially yielding promising results with minimal fine-tuning and reducing the overall trainable parameters.

2. Related Work

CLIP (Radford et al., 2021) aligns images and text in a shared space, enabling zero-shot learning. LLMs like GPT-3 and Llama-3 excel in generating contextually appropriate text but are primarily designed for textual data. ClipCap[13] (Mokady et al., 2021) uses a mapping network to transform CLIP image embeddings for image captioning. Unlike ClipCap, our method integrates both image and prompt embeddings into the LLM, enhancing tasks beyond captioning, such as image-question answering. Additionally, our combined Cross-Entropy (CE) and Self-Critical Sequence Training (SCST) loss function optimizes for both token-level accuracy and sequence-level quality. SCST (Rennie et al., 2017) improves sequence-level performance by optimizing based on rewards from evaluation metrics, promoting coherent and contextually relevant text generation.

3. Methods

Our approach includes a new architecture that combines CLIP and LLMs, a fine-tuning technique for the LLM, and a reinforcement learning-based training objective to optimize sequence-level performance.

Model Architecture:

Our proposed model architecture leverages the strengths of both CLIP and a pre-trained LLM (such as Llama-3 or GPT-3). The architecture consists of three main components:

1. CLIP Model: Used to extract rich semantic embeddings from images.
2. Mapping Network: A simple Multi-Layer Perceptron (MLP) that projects CLIP embeddings into the LLM embedding space.
3. LLM: Processes both textual prompts and image embeddings to generate meaningful responses.

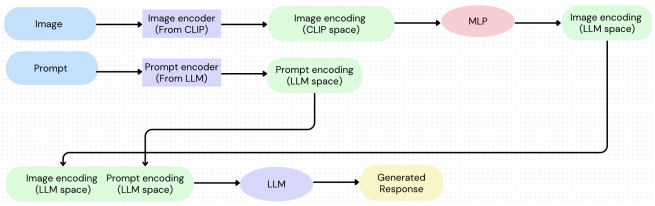


Figure 1: model architecture

Training Pipeline

We propose two training approaches for the model: jointly training the mapping network while fine-tuning the LLM, or freezing the LLM and solely training the mapping network. Our objective is to train the model on image-caption pairs and evaluate its generalization to tasks like image question answering. To achieve this, we will utilize a dataset comprising 4 million randomly chosen data points from the LAION-400[4] dataset, consisting of images and their captions. Additionally, we will introduce a novel loss function based on reinforcement learning techniques to improve the model's ability to generate realistic responses. If we choose to train the mapping network and fine-tune the LLM jointly, these are the steps we would follow:

1. Initialize the pre-trained Llama-3 and ViT-G/14 models.
2. Extract image embeddings using the ViT-G/14 model.
3. Project these embeddings into the LLM space using the mapping network (MLP).
4. Encode text prompts using Llama-3 and concatenate these with the projected image embeddings.
5. Use the combined embeddings as input to Llama-3 to generate responses.
6. compute the loss using the new loss function (defined below)
7. Perform backpropagation and update both the mapping network and LLM weights.

If we only wanted to train the mapping network, we would freeze the weights of the LLM and in step 7 we would only update the weights of the mapping network.

Loss function:

Our training objective combines two components: Cross-Entropy (CE) Loss[10] and Self-Critical Sequence Training (SCST) Loss. This combined loss function aims to optimize both token-level prediction accuracy and sequence-level performance, which is particularly important for generating coherent and contextually relevant responses.

Cross entropy loss:

The Cross-Entropy Loss is a standard loss function used in language models for token-level prediction. It measures the difference between the predicted probability distribution of the next token and the actual distribution (one-hot encoded ground truth).

Mathematical Formulation:

The CE loss is defined as:

$$\mathcal{L}_{ce} = -\sum_{t=1}^{T} \log P(y_t|\mathbf{y}_{<t}, \mathbf{x})$$

Where $y = (y_1, y_2, ..., y_T)$ is the sequence of ground truth tokens, $\widehat{y} = (\widehat{y_1}, \widehat{y_2}, ..., \widehat{y_T})$ is the sequence of predicted tokens, and $P(y_T | y_{<t}, x)$ is the probability of the token y_t given the previous token in the input (image and prompt).

Self-Critical Sequence Training (SCST) Loss:

The SCST Loss is a reinforcement learning-based objective that optimizes for sequence-level metrics directly. SCST uses the difference in rewards between sampled and greedy decoded sequences to update the model, promoting sequences with higher metric scores.

The SCST loss is defined as:

$$\mathcal{L}_{pg} = -\left(R(\mathbf{y}_{\text{sampled}}) - R(\mathbf{y}_{\text{greedy}})\right) \log P(\mathbf{y}_{\text{sampled}}|\mathbf{x})$$

Where y_{greedy} is the sequence generated by greedy decoding, $y_{sampled}$ is the sequence generated by sampling and $R(y)$ is the reward function.

For the SCST loss, we'll employ the Metric for Evaluation of Translation with Explicit ORdering (METEOR)[3] reward function. Unlike Bilingual Evaluation Understudy (BLEU)[6], METEOR incorporates synonymy, stemming, and flexible alignment strategies to evaluate text quality comprehensively. Its features include exact matching, stemming, synonymy, and alignment, while penalizing fragmented and out-of-order matches to enhance fluency.

Combined Loss Function:

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda \mathcal{L}_{pg}$$

Where λ is a hyperparameter that controls the trade-off between the CE loss and the SCST loss.

The combined loss function integrates CE loss and SCST loss to optimize both token-level accuracy and sequence-level quality. CE loss ensures grammatical correctness and fluency by aligning token predictions with ground truth sequences. SCST loss, using reinforcement learning, improves sequence-level metrics like CIDEr and BLEU, fostering coherent, contextually accurate, and diverse outputs by exploring different sequences. This dual approach enables robust language generation, balancing syntactic correctness and contextual relevance, and allows fine-tuning for specific task requirements, enhancing performance across diverse tasks and datasets.

Code for training linked here: Colab notebook (Use Caltech Email)

4. Experiments

Our experiment has 2 main goals: to evaluate Model Performance - Assess the ability of the proposed method to generate meaningful responses to textual prompts based on both textual and image inputs. This involves measuring the quality, coherence, and relevance of the generated text in relation to the provided prompts and images and to evaluate the model's generalization Across Tasks - Investigate the generalization capabilities of the model across diverse multimodal tasks, such as image captioning, image question answering, and text-based image retrieval. We aim to demonstrate the versatility of the proposed method in effectively addressing various multimodal comprehension tasks.

Data Selection and Access

We select the LAION-400[4] dataset as the primary dataset for our experiments. LAION-400 comprises a diverse collection of images paired with descriptive captions. Access to the LAION-400 dataset is facilitated through the LAION platform (https://laion.ai/blog/laion-400-open-dataset/), ensuring ease of retrieval and preprocessing.

Model Selection and Access

We will use the Llama-3 [1] model as our LLM. This is because Llama-3 has been pre-trained on a large corpus of diverse text data, enabling it to capture intricate linguistic patterns and semantic nuances, leading to very generations with high coherence and relevance to the prompt. In addition, it is open - source and can be accessed by anyone. We will also use the ViT-G/14 model from open_clip[5] as our image encoder from CLIP.

Experiment 1: Evaluate Model Performance

Objective: Assess the ability of the proposed method to generate meaningful responses to textual prompts based on both textual and image inputs. This involves measuring the quality, coherence, and relevance of the generated text in relation to the provided prompts and images.

Method: We train the proposed model on the LAION-400 dataset using a joint training approach, simultaneously training the mapping network and fine-tuning the LLM. For each image in the LAION-400 dataset, we randomly select a textual prompt from a predefined list of prompts, given below. These prompts cover a range of topics and styles, ensuring diversity in the input provided to the model. We will have to experiment a little bit to find the best set of prompts to use. We will do some hyperparameter tuning to find the best number of hidden layers to use for the MLP and the best value of λ. During evaluation, we provide a randomly chosen textual prompt along with an image from a different dataset to the trained model and measure the quality of the generated responses using standard evaluation metrics such as BLEU, Consensus-based Image Description Evaluation (CIDEr)[7], Visual-semantic Matching-based Description (VMD)[8] and Semantic Propositional Image Captioning Evaluation (SPICE)[9]. Additionally, we conduct human evaluation to assess coherence and relevance by asking the user if this was a good response or not.

Experiment 2: Generalization Across tasks

Objective: Investigate the generalization capabilities of the model across diverse multimodal tasks, including image captioning, image question answering, and text-based image retrieval. We aim to demonstrate the versatility of the proposed method in effectively addressing various multimodal comprehension tasks.

Method: Similar to experiment 1, we train the proposed model on the LAION-400 dataset using a joint training approach, simultaneously training the mapping network and fine-tuning the LLM. For each image in the LAION-400 dataset, we randomly select a textual prompt from a predefined list of prompts, given below. These prompts cover a range of topics and styles, ensuring diversity in the input provided to the model. We will do some hyperparameter tuning to find the best number of hidden layers to use for the MLP and the best value of λ. During evaluation, we

provide the model with images from the VQA dataset along with their corresponding questions from the dataset. The model processes both the visual information from the image and the textual information from the questions to generate answers. These answers are then compared against the ground truth answers provided in the dataset. We use evaluation metrics such as BLEU, CIDEr, VMD and SPICE on the generated response, comparing these to the ground truth answers. We will also ask humans to indicate if this was a good response or not.

Predefined list of prompts for training

Describe this image in a small sentence, Describe this image concisely, Describe what you see in this image in less than 12 words, What does this image show? Describe your understanding of the image in less than 12 words, Describe the image in simple words.

We will repeat experiments 1 and 2, training only the MLP and not the LLM. Additionally, we will evaluate GPT-4 using the methods outlined in these experiments. Mock figures for the results display of experiments 1 and 2 are shown in Figures 2 and 3.

5. Discussion

Positive results from our experiment would signify the effectiveness of our method in generating meaningful responses to textual prompts based on both textual and image inputs. Comparable scores to GPT-4 in evaluation metrics such as BLEU, CIDEr, VMD, and SPICE, coupled with positive human evaluation scores for coherence and relevance, would validate the utility of our approach. These outcomes would demonstrate that our method is a valuable tool for addressing multimodal comprehension tasks Additionally, positive outcomes would mean the model generalizes well to tasks it was not trained to do. This would mean comparable scores to GPT-4 in evaluation metrics like BLEU, CIDEr, VMD and SPICE for image question answering tasks. generalization across diverse multimodal tasks. These positive outcomes would mean that our model, with minimal training, and only fine tuning, can achieve the comparable results to GPT-4. Negative results may suggest limitations in our method, possibly stemming from factors like inappropriate predefined prompts, suboptimal reward functions, MLP complexity issues, or overfitting of the LLM during fine-tuning. To address these challenges, potential solutions include freezing the LLM after a certain training threshold, adjusting the MLP complexity, refining the prompt selection process using prompt engineering, and modifying the reward function for our training objective, such as adopting the CIDEr score instead. Low evaluation metric scores may indicate issues with prompt selection or model architecture, while discrepancies between model-generated responses and human evaluations could suggest shortcomings in the reward function. Overfitting of the MLP will be evident if the model performs well on training data but poorly on unseen data in experiments 1 and 2. Similarly, overfitting of the LLM will be observed in experiment 2 if the model fails to generalize well to image question answering tasks after excelling in experiment 1.

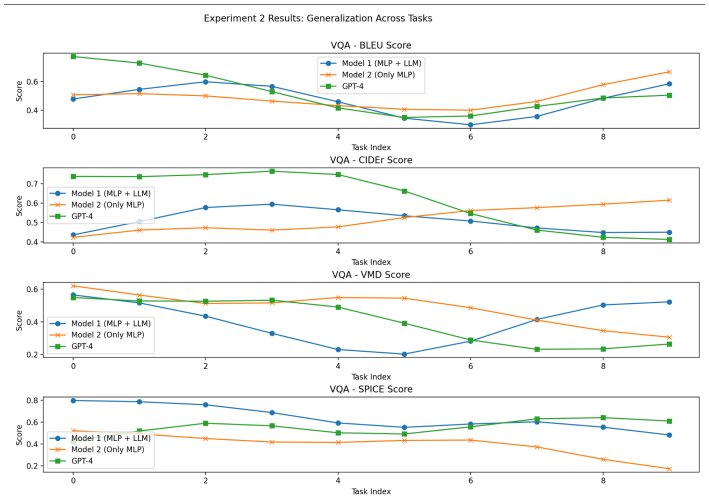


Figure 3: mock plot of experiment 2. Task index refers to the prompt

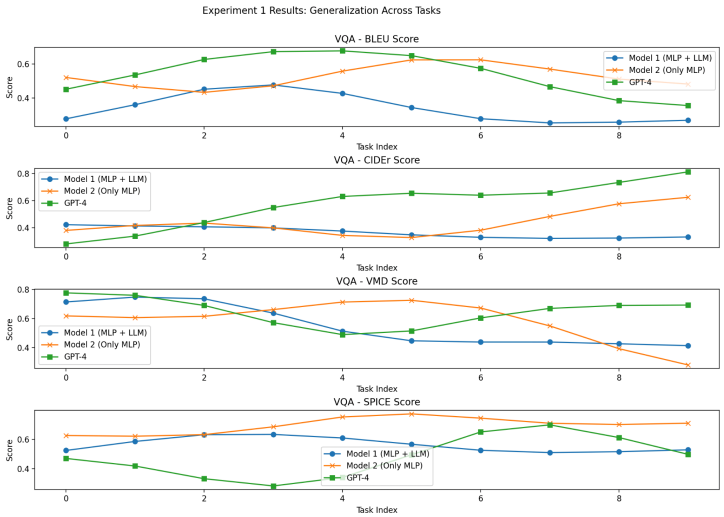


Figure 2: mock plot of experiment 1. Task index refers to the prompt

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